

SAINT HUBERT AB, Belgium

WMO: 064760

Lat: 50.0389N

Lon: 5.4039E

Elev: 588

StdP: 94.45

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.1	-13.6	1.2	-5.8	-10.6	1.6	-4.8	9.3	2.8	8.5	2.1	3.6	50	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	26.7	18.6	24.5	17.6	22.8	16.4	19.3	25.0	18.2	23.3	17.2	21.6	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.2	21.6	16.2	12.4	20.4	15.4	11.7	19.5	57.5	25.1	53.7	23.5	50.6	21.6	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.2	6.4	-11.1	30.4	3.0	2.0	-13.3	31.9	-15.0	33.1	-16.7	34.2	-18.8	35.7
			-11.5	21.4	2.9	1.1	-13.6	22.2	-15.3	22.8	-16.9	23.4	-19.0	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	0.1	0.6	4.2	8.0	11.0	14.1	16.3	15.6	12.2	8.7	4.5	1.6
	DBStd	6.71	3.70	4.30	3.99	4.09	3.92	3.53	3.67	3.26	3.07	3.76	3.53	3.65
	HDD10.0	1387	308	263	184	87	35	3	0	1	10	68	168	261
	HDD18.3	3778	566	496	439	311	229	136	87	98	186	297	415	519
	CDD10.0	699	0	1	3	26	66	126	194	176	77	29	2	0
	CDD18.3	48	0	0	0	0	2	8	22	14	2	0	0	0
	CDH23.3	324	0	0	0	0	8	61	153	94	7	0	0	0
	CDH26.7	69	0	0	0	0	0	9	38	21	0	0	0	0
Wind	WSAvg	3.4	3.8	3.7	3.9	3.3	3.3	3.2	2.9	2.8	3.0	3.4	3.6	3.8
Precipitation	PrecAvg	1040	107	87	78	62	74	79	88	97	77	84	90	117
	PrecMax	1290	269	209	169	161	171	161	211	241	166	196	151	277
	PrecMin	763	12	13	11	5	30	20	42	23	13	36	5	18
	PrecStd	138	51	43	42	37	35	37	37	56	45	32	37	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.9	16.8	21.5	24.4	28.0	30.5	29.2	24.3	20.3	14.4	10.1
		MCWB	8.6	5.5	9.3	12.9	17.3	19.2	19.3	19.2	17.4	14.3	12.6	9.0
	2%	DB	7.9	9.1	14.6	19.2	21.7	25.0	27.0	25.9	21.3	17.7	12.0	8.8
		MCWB	7.4	5.0	8.2	11.6	14.6	17.7	18.5	18.3	15.9	14.2	10.5	7.8
	5%	DB	6.3	7.1	12.0	16.8	19.8	22.7	24.6	23.5	19.3	15.4	10.7	7.7
		MCWB	5.8	5.4	7.3	10.8	13.8	16.4	17.5	17.3	15.0	12.8	9.6	6.8
	10%	DB	5.0	5.9	9.6	14.6	17.7	20.7	22.8	21.2	17.2	13.9	9.3	6.4
		MCWB	4.5	4.7	6.5	9.7	12.9	15.5	16.6	16.1	13.7	11.9	8.4	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	7.7	10.1	14.0	18.2	20.5	20.5	20.6	18.1	15.6	12.9	9.5
		MCDB	8.9	9.9	14.1	20.2	23.4	26.8	28.3	26.3	23.4	18.3	14.1	9.9
	2%	WB	7.4	6.6	9.0	12.2	16.3	18.5	19.3	18.8	16.5	14.5	10.7	8.1
		MCDB	7.8	7.8	13.3	17.6	19.6	23.5	25.4	24.1	20.2	16.9	11.7	8.7
	5%	WB	6.0	5.7	7.9	11.1	14.6	17.2	18.3	17.7	15.4	13.4	9.6	7.0
		MCDB	6.3	6.8	10.8	16.1	18.4	21.5	23.6	22.4	18.7	15.3	10.6	7.6
	10%	WB	4.6	4.7	7.0	10.1	13.2	16.1	17.2	16.7	14.3	12.2	8.5	5.9
		MCDB	5.0	5.9	9.5	14.4	17.0	20.0	21.8	20.6	16.8	13.8	9.4	6.4
Mean Daily Temperature Range	5% DB	MDBR	3.6	4.5	6.5	7.9	8.1	8.2	8.3	7.8	7.1	5.5	4.0	3.7
		MCDBR	4.3	6.7	10.2	10.5	10.7	11.1	10.9	10.5	9.4	7.6	5.7	4.6
		MCWBR	4.1	4.7	5.5	5.7	5.9	5.7	5.1	5.4	5.3	4.8	4.4	4.1
	5% WB	MCDBR	4.2	5.7	8.2	10.0	9.2	10.0	9.9	9.2	8.4	6.7	5.1	4.3
		MCWBR	4.2	4.4	5.2	5.7	5.6	5.6	5.1	5.4	5.0	4.5	4.2	4.2
Clear-Sky Solar Irradiance	taub	0.293	0.316	0.345	0.374	0.383	0.384	0.383	0.377	0.350	0.335	0.311	0.291	
	taud	2.462	2.421	2.361	2.300	2.305	2.319	2.314	2.356	2.416	2.463	2.472	2.440	
	Ebn at Noon	762	828	859	868	874	874	868	856	844	793	729	704	
	Edh at Noon	60	79	100	118	123	122	121	111	94	75	59	53	
All-Sky Solar Radiation	RadAvg	0.80	1.50	2.67	4.19	4.96	5.46	5.20	4.43	3.37	1.97	0.93	0.63	
	RadStd	0.12	0.28	0.37	0.51	0.53	0.50	0.57	0.49	0.35	0.17	0.11	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page